

Putnam POTD November 18, 2025

I. PROBLEM

(1990 A3) Prove that any convex pentagon whose vertices (no three of which are collinear) have integer coordinates must have area greater than or equal to $5/2$.

II. SOLUTION

Recall from Pick's theorem that for a convex polygon P , Let I be the set of grid points inside P , and B be the set of grid points on the boundary of P , then

$$\text{Area}(P) = |I| + \frac{1}{2}|B| - 1.$$

In our case, $B = 5$, thus we only need to show that $I + \frac{3}{2} \geq \frac{5}{2}$, so we need to show $I \geq 1$.

Since all coordinates are integers, they can have four possible parity

configurations: $(E, E), (O, O), (O, E), (E, O)$, which can be seen as 4 pigeonholes, and since there are 5 points, so there must be two points with the same configuration. For two points with the same configuration, their midpoint must be a grid point.

There are two cases: If the two points are diagonally across from each other, their grid point is in the interior, then there is a grid point inside and we are done.

Otherwise, the two points are on the boundary, and their midpoint is also on the boundary, which reduces the problem to a smaller subproblem, as we have another convex pentagon on grid points again.

From a high level perspective, this process will terminate eventually, hence there will always be an interior point, and thus $\text{Area}(P) \geq \frac{5}{2}$. In fact, the construction for $\text{Area}(P) = \frac{5}{2}$ is given by the polygon $(0, 0), (1, 0), (0, 1), (1, 1), (1, 2)$.